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EDITORIAL

Deep Decarbonization Pathways for national development consistent with carbon neutrality

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ABSTRACT
The global carbon neutrality goal set by the Paris Climate Agreement requires socially, economically and politically actionable strategies to reach net zero greenhouse gas (GHG) emissions at national levels. These strategies must be informed by robust, iteratively updated evidence capturing the specifics of diverse national circumstances and developed with all key stakeholders in a consultative fashion. The Deep Decarbonization Pathways (DDP) methodology provides an open but structured means to elaborate, communicate and debate such country-driven scenarios of long-term transitions. This special issue presents advancements in the DDP methodology to integrate key features required to investigate carbon neutrality. This includes the detailed representation of the Agriculture, Forest and Other Land Uses (AFOLU) sector as a key source of negative emissions; a detailed sectoral approach across transport, industry, land-use and power generation that considers a broad range of mitigation options; and the explicit investigation of the compatibility with socio-economic development priorities as defined by each country. The scenario analysis conducted in Argentina, Brazil, China, India, Indonesia, Mexico and South Africa highlights the necessity for wide-ranging sector-focused policy packages supported by adequate international cooperation, in order to simultaneously address multiple objectives in a consistent policy framework. The detailed policy insights gathered in this Special Issue provide concrete benchmarks against which progress of countries towards carbon neutrality can be assessed.

KEYWORDS
Carbon neutrality; deep decarbonization; national development; pathways analysis; policy packages; sector-focused

Introduction

Formation of policies to reach emissions levels consistent with the Paris Agreement goal requires a detailed understanding across stakeholders of potential transformation pathways between how sectors operate today given country contexts and how they could in a low-GHG emissions future.

The low GHG emissions scenarios literature has historically been dominated by global-scale top-down approaches, which typically derive global transformations using cumulative emissions constraints defined by carbon budgets (van Beek et al., 2020; Weyant, 2017). Using this approach, country-scale insights often emerge via downscaling from global scenarios. Although useful to put national transitions in the global context, this approach fails to capture national circumstances, required by national decisionmakers to be understandable and implementable for shaping and negotiating national policy (Gambhir et al., 2022). In response to the adoption of a nationally focused paradigm for climate action in the Paris Agreement, a body of literature

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adopting a bottom-up perspective, centred on countries and regions with the power to implement policy packages, has grown rapidly over the last decade (Lecocq et al., 2023).

The Deep Decarbonization Pathways (DDP)[1] has been a flagship initiative represented in this body of literature since its launch in 2013. The papers in this Special Issue present the latest methodological advancements within this research initiative and the resulting research and policy insights informing national transitions towards carbon neutrality.

The first section of this editorial presents the main innovative features of the DDP approach to national scenario design, which define the starting point of the work conducted in this SI. The following section highlights the key methodological improvements implemented in the papers of this Special Issue to make the DDP approach relevant to explore carbon neutrality. We then discuss the main policy messages emerging from a cross-cutting analysis of the studies conducted in this Special Issue. The final section concludes by connecting this research with on-going policy processes and identifying research gaps to be explored in future research.

An innovative method for policy-relevant low carbon national pathways

The DDP has promoted innovative approaches and methods to address core research questions at the national level (Bataille et al., 2016):

- How can countries contribute to the global climate agenda without relying on a top-down 'burden sharing' approach which defines country actions as a global allocation problem?
- What are the short-term actions and policies that are required to put and keep a country on track with its long-term (mid-century and beyond) objectives?
- How can a country align its broad security, development, mitigation and adaptation objectives?

Key methodological advancements promoted by the DDP to address these research questions include:

- **the empowerment of in-country research teams**, who are free to choose their models and assumptions according to their national policy priorities, systems' characteristics and capacity constraints (Pye & Bataille, 2016). Working from their independent national perspectives, these national partners operate as a community of practice learning from each other and elaborating cross-country messages of global relevance from the composite of the national studies.
- **the adoption of a common structured reporting template for national pathways, combining quantitative outputs and narratives**. These elements focus on the detailed content of transformations at the sectoral level and capture a wide range of mitigation options, drivers, and co-benefits beyond those that may be represented in models, including non-technological and organizational options. This approach crucially puts emissions pathways in the broader context of underlying development pathways (Winkler et al., 2015) and provides evidence for relevant actions and instruments to trigger concrete transformations at the national and international scales (IDDRI, 2021; Kuramochi et al., 2018; Pauw et al., 2018).
- **the adoption of a 'backcasting from the goal' approach** (in contrast with conventional 'forecasting'), which focuses on exploring ways to reach set long-term objectives from the present (Robinson, 1982). This approach is implemented through an iterative assessment of progress, options, and policies towards the desired sectoral transformations guided by updating, comparison, benchmarking and adaptive management (Kaufman & Bataille, 2025).
- **the recognition of deep uncertainty**, which forces analysts to go beyond conventional parametric uncertainty and rather consider scenarios distinguished by structural differences mapping a wide range of possible futures (Maier et al., 2016; Mathy et al., 2016; Morgon & Henrion, 1990). This approach improves understanding of constraints, trade-offs, synergies and choices on the pathways to carbon neutrality (Raskin & Swart, 2020; Schweizer & Kriegler, 2012).

[1]https://ddpinitiative.org/

These components are articulated in a specific pathways design framework, which brings together narratives, models and reporting in an iterative scenario design process (Waisman et al., 2019). This approach to scenario design is suited to support a structured, ideally iterative, engagement with real-world decision makers and other stakeholders, informed by scientifically robust analysis using non-technical narratives. It can inform the formation of policy packages, and therefore help guide decision making in a context of deep uncertainty and challenging political economy issues related to vested interests (Kane & Boulle, 2018).

Methodological advancements from this Special Issue

The operationalization of the above DDP methodological principles has followed the evolution of international policy conversations on climate, which has led to functional modifications.

Initially grounded on the exploration of the 2°C global temperature goal during the 2013–2015 period, when this was the scientific target anchoring most ambitious scenarios, the work of the DDP has shifted to consider the 'well below 2°C and towards 1.5°C' objective codified by the Paris Agreement. Following the increasing number of carbon neutrality targets taken by countries, this has led the DDP to investigate the conditions for countries to reach net zero CO_2 and then GHG emissions by mid-century or soon after (Briand et al., 2026). These changes in climate policy ambition require changes to the practical conduct of the pathways analysis.

First, investigating carbon neutrality puts emphasis on natural carbon flows to and from the agriculture and land-use sector. It requires specific analysis of all related drivers and constraints affecting this sector at the national level, such as land-use competition, changes in agricultural practices, and trade-offs with biodiversity or socio-economic impacts. It also requires explicit consideration of key interactions between the AFOLU sector and other economic sectors e.g. fertilizer and fuel needs for the agricultural sector are produced by the industrial sector, or biomass is needed for the energy production, transport, building and industrial sectors. In this Special Issue, building on Svensson et al. (2021), Svensson et al. (2024) describe how approaches adopted by research teams in Brazil, India and Indonesia focus on six key transformation drivers: agricultural production and management practices; the socio-economic organization of farms; forestry and forests-related management practices; demand for AFOLU products, including diets, trade and waste; and key elements regarding land ownership and occupation (urbanization trends, land tenure, etc.).

Second, the depth of the changes required to reach carbon neutrality requires the adoption of a detailed sectoral approach that considers a broad range of mitigation options. In addition to technological options conventionally considered in modelling studies, this means including structural and organizational changes, with a particular focus on ambitious efforts towards reductions of energy and material demand in each sector. Briand et al. (2024) describe and apply a pathways design framework for freight transport which includes five key groups of sectoral decarbonization drivers: (i) the future demographic, economic, spatial and socio-cultural structure of consumption, production and trade, (ii) the development and management of transport and logistics infrastructure, (iii) the development of vehicles and truck technologies and their market penetration, (iv) the organization of logistics operations (supply and delivery), modal and vehicle choices and (v) the production and distribution of fuels. Building on Lefèvre et al. (2020), Briand et al. (2023) similarly apply a framework distinguishing eight categories of transition driver for passenger transport: demography and economy; human settlement, land development and spatial organization; sociocultural practices and lifestyles; technological development of vehicles; fuel production and carbon content; penetration of alternative engine types and fuels; income dedicated to transport, modal distribution and costs; and speed, infrastructure and time dedicated to transport. Bataille et al. (2023) describe a methodology for designing decarbonization pathways for steel and cement use and production which integrates demand and supply side drivers and policies: (i) evolution of demand for each product, including those coming from targeted policy action, (ii) material efficiency that uses less material in the construction and design of infrastructure, (iii) product substitution enabling use of less GHG intensive material for the same demand, (iv) specific action on industrial process emissions.

Finally, to be policy-relevant, carbon neutrality analyses need to demonstrate explicitly the compatibility of the required deep transformations in all sectors with socio-economic development priorities as defined by each country. In practice, this means that pathways analyses must go beyond the aggregate representation of socio-economic dimensions, conventionally limited to economic growth as measured by GDP, and instead have more explicit socio-economic narratives relevant to each country context. Briand et al. (2026) describe how the pathways design method integrates the priority national socio-economic objectives at the core of the analysis, such as reduction of energy insecurity and inequalities, and improved access and employment. The sectoral analyses conducted in the other papers of the special issue also highlight sector-specific socio-economic parameters. Svensson et al. (2024) explore how profound transformations of the AFOLU sector can be achieved consistently with other AFOLU environmental and socio-economic goals, such as adapting to climate change, conserving biodiversity, ensuring food security and improving rural livelihoods. The passenger transport analysis conducted in Briand et al. (2023) focuses on issues like the quality of service proposed by different transport modes, sociocultural practices and lifestyles, income and time dedicated to transport, acknowledging the different types of mobility and the associated constraints (notably daily commuting). Briand et al. (2024) stress the interplay between freight transport, industrial organization and consumers' preferences. Bataille et al. (2023) highlight the structural connection between cement, steel and infrastructure deployment as required to support national development.

Carbon neutrality requires wide ranging sector-focused policy packages supported by adequate international cooperation

By adopting the innovative methodology described in the previous section, this Special Issue offers specific policy insights grounded on original research on carbon neutrality at the national level, with a focus on large emerging economies and sectors previously unexplored in detail.

The carbon neutrality transition is about addressing multiple objectives in a consistent policy framework. Indeed, carbon neutrality requires accelerating immediate emission reductions in sectors where technical solutions are already available (notably power generation, passenger transport and AFOLU), while preparing conditions for long-term deep emission reductions. The latter requires proactive and anticipatory actions on technological progress, infrastructure and behavioural change. Carbon neutrality also requires strengthening synergies between climate, industrial and economic agendas and managing the social impacts of the transition, notably with regards to the most vulnerable. From a policy standpoint, addressing these multiple objectives in a balanced way calls for the adoption of comprehensive policy packages that combine complementary actions in a consistent policy framework (Briand et al., 2026).

The analysis conducted in this Special Issue provides concrete examples of such a holistic approach to policy and stresses the necessity to approach policy design in a way that captures the specificities of each of the key sectors that matter for deep decarbonization. It investigates some blind spots in analyses of sectors which are most frequently considered in the decarbonization literature, i.e. power generation and passenger transport. In addition, a core addition of this Special Issue is a structured approach to the analysis of other sectors, which have remained less considered in decarbonization analyses (i.e. energy-intensive industries, freight transport, AFOLU).

To achieve power sector decarbonization, a step change increase in type of policy and policy stringency is required, especially to provide sufficient firm clean electricity to support deep uptake of intermittent renewables (Bataille et al., 2021). This includes considering necessary reforms to make these markets self-sustaining and to enable investment to meet development, energy security and climate goals. In power generation, the escalation of stranded assets (e.g. coal mines, power plants, jobs, communities) required to reach the most ambitious climate goals raises the need for targeted actions in favour of just and equitable transitions, as well as well-designed international financial support through mechanisms like Just Energy Transition Partnerships (Vishwanathan et al., 2025).

In passenger transport, carbon neutrality requires going beyond the conventional approach focused on the diffusion of low-carbon technologies, notably electric vehicles. To achieve deep emission reductions, a wide spectrum of mitigation options should be considered, including systemic measures related to demand-side options. These options include targeted land-use, social, and urban policies which could reduce distances between activities, the costs of mobility and time, thereby improving quality of life and supporting the shift to non-motorized and public transport (Briand et al., 2023).

In steel and cement, half of GHG emissions can be mitigated through simple, inexpensive and already commercialized actions, including more high-quality recycling of metals and maximizing cementitious material substitutes. (Bataille et al., 2023) highlight the importance of organizational and cultural changes, notably the concentration of production of cement and concrete in professional facilities in developing countries, to facilitate the use of available techniques. But they also show that deeper emission reductions, as needed for carbon neutrality, require technology development and financial support through dedicated investments (notably in low emission primary iron reduction and mixed primary and secondary steel making), lead markets to facilitate deployment of very low GHG product in the early phases where cost competitiveness may not be guaranteed, and financing from developed countries. In addition, the vast informal construction sector must be engaged in a way that it can absorb and use the mitigation strategies, e.g. the use of substitute cementitious materials in the place of clinker in cement mixes. The use of smartphone communications, QR code markings, etc. to communicate key use data, and the development of informal norms will be key to transforming these under-monitored and under-regulated sectors. International cooperation and support are needed to agree on and implement CO_2 intensity accounting systems, improve access to low-emissions production technologies through innovation and commercialization combined with technology transfer or co-development, and offer financial and technical support for clean production investments. (Otto & Oberthür, 2024) highlight global governance as a critical lever to enable the sectoral transformations of energy-intensive industries required by carbon neutrality, but its potential has been so far underexploited. Key governance gaps, including rules for market creation and carbon leakage, means of implementation and inter-institutional coordination, are more rooted in geopolitics rather than a lack of suitable institutions

In the freight transport sector, the scenario analysis stresses the importance of structural and systemic mitigation options that influence industrial production and supply chain structure, as well as modal and logistics choices, beyond the technological options related to road freight vehicles and fuels. The implementation of such actions could drastically reduce emissions per tonne-km by 2050, while also reducing energy consumption and supporting national development (Briand et al., 2024). More specifically, a diversity of economic policy tools can play a critical role in freight transport decarbonization, such as direct national government investments to improve transport and storage infrastructure, public-private partnerships, and green bonds. However, economic instruments alone are insufficient to cover policy needs when it comes to the management and maintenance of infrastructure, which are key areas for sectoral decarbonization; for this, non-economic and regulatory policies combined with strategic investments in infrastructure and planning could offer a large set of options to manage efficiently access to the infrastructure access. These actions notably depend on the availability of international financing for low-carbon infrastructure as a key global enabling condition (Harry-Villain et al., 2025).

Finally, in agriculture, forest and other land uses (AFOLU), a key challenge is to design policies that maximize co-benefits and avoid trade-offs between environmental and socio-economic objectives, alongside the challenges of policy implementation. The analysis identifies the need to prioritize measures that support the restoration of degraded land for agriculture and/or forests, increase integrated land uses, such as agroforestry, sustainably increase crop- and livestock productivity, and rapidly reduce deforestation. Concrete policy instruments include economic incentives to land managers for conservation or more environmental land management practices, changing land use regulations to protect ecosystems and/or encourage shifts in land management practices and strengthening enforcement capacity of land protection through, e.g. improved measurement, reporting and verification systems. In the long run, national policies, supported by global measures, will require the AFOLU sector to become as net-negative as possible given national constraints (Svensson et al., 2024)

Conclusion

In this Special Issue we have explored recent contributions to the bottom-up modelling and scenario literature to inform policy relevant transition pathways and policy packages consistent with carbon neutrality, especially in developing country contexts. The papers in this Special Issue describe how to address some of the critical methodological challenges posed by the depth of transformations required by the carbon neutrality objective, including the need for sectoral granularity and technology detail, and inclusion of a broad range of mitigation options. The analyses stress the need for comprehensive policy packages, anchored in the specificities of each sector and supported by international cooperation, in order to simultaneously address multiple objectives in a consistent policy framework. The detailed policy insights gathered in this Special Issue provide concrete benchmarks against which progress of countries towards carbon neutrality can be assessed. As such, this analysis can help identify priorities for enhancement of national climate action.

A recent analysis on the last decade of national climate action highlights already concrete policy gaps with respect to key policy insights from this Special Issue on the requirements for carbon neutrality (DDP, 2025). Notably, this assessment shows that too little proactive action has been taken to prepare for the longer-term evolution of infrastructure and behaviours needed to reach carbon neutrality. Also, despite growing recognition of the close links between climate and social priorities, and the need to articulate climate policies with industrial, trade and other socio-economic policies, the emergence of consistent policy packages coherently and consistently addressing climate along with social and industrial concerns is still largely missing.

These two critical gaps highlight the necessity to anchor the design of policy at national level on a robust, evidence-based and widely accepted national strategy, which can provide guidance to short-term decisions within the broad context of sustained longer term economic development. Sectoral and economy-wide policies would indeed benefit from being guided by targeted yet adaptable strategies and roadmaps, to ensure that policy actions are defined according to a clear vision of the technological and investment transitions they should trigger. The carbon neutrality transition, therefore, requires the establishment of innovative governance organizing inclusive processes supported by strong mandates, explicitly framed by a long-term perspective and informed by country-driven scientific evidence.

Building on advancements described in this Special Issue, ongoing, iterative development of the scientific literature on national pathways to carbon neutrality will be needed to guide sectoral transitions, including their drivers and mitigation options. Another critical priority for further research should be to deepen the understanding of the social and economic implications of national carbon neutrality transitions to inform the necessary conditions of the alignment between climate ambition and industrial, economic and social priorities as defined at the national level.

These analyses will be a core foundation to guide concrete implementation of ambitious climate action at the national level in the coming years, to inform the process of stocktaking on climate action at the global level in the lead up to the 2028 Global Stocktake of the UNFCCC and to help design more ambitious national plans and strategies.

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